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Question Paper Code : 21576

B.E. / B.TECH. DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2024

Third Semester

Computer Science and Engineering

23CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2023)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

A, CO1 1. Add +8 and -5 using 2's complement method.

U, CO1 2. The instruction Load R2, NUM1 belongs to which addressing mode? Justify your answer.

A) Immediate Addressing B) Absolute Addressing

C) Memory Addressing D) Register Addressing

A, CO2 3. Show the diagram for implementing 2-port register file using single port register files.

U, CO2 4. Write the assembly language notation equivalent to the RTN: R3 <- [LOC], RTN: R3 <- [R1] + [R2].

U, CO3 5. What do you mean by RAW dependency? Give one example.

U, CO3 6. A four-stage pipeline has stage delays of 100,110,120 and 105 nanoseconds. The registers that are used between the pipeline stages have a delay of 10 nanoseconds each. The total time to execute 200 independent instructions on this pipeline, assuming there are no pipeline stalls, is _____ nanoseconds. Justify your answer.

A) 25240 B) 26390 C) 24010 D) 20010

U, CO4 7. In _____ mechanism both the cache location and the main memory location are updated simultaneously. Justify your answer.

A) Write-through B) Write-back C) Copy-back D) Load-through

A, CO4 8. _____ number of 256 K × 16 chips are required for creating 16 M × 64 memory and _____ number of address bits are required to address it. Justify your answer.

A) 256, 64 B) 256, 24 C) 128, 64 D) 128, 24

U, CO5 9. It is important to clear the input status bit in an interface circuit as soon as the input data register is read. Why?

U, CO5 10. _____ interrupt can be temporarily ignored by the counter. Justify your answer.

A) Vectored interrupt B) Maskable interrupt

C) Non-maskable interrupt D) Low priority interrupt

PART B – (4 × 15 = 60 Marks)

A,CO2 11. a) i. Write the sequence of actions needed to fetch and execute the following instruction using datapath register:

A) ADD R1, R2, #100

B) Store R4, X(R5) (8 Marks)

U,CO2 11. a) ii. With a neat diagram, show how control signals are generated using hardwired control. (7 Marks)

OR

A,CO2 11. b) i. Draw the datapath of a processor with 5 stages of execution for the Load R1, X(R2) instruction. Also show the fetch and execution steps for the above instruction. (8 Marks)

U,CO2 11. b) ii. Explain in detail about hardware organization required for implementing the CISC processors. (7 Marks)

A,CO3 12. a) i. Identify if any hazards exist in the following code:

ADD R1, R2, R3

SUB R4, R1, R2

AND R5, R4, R1

If any hazards exist, suggest an alternative solution and justify your answer with necessary pipeline timing diagram. (8 Marks)

U,CO3 12. a) ii. Write detail notes about different branch prediction techniques in pipelining to avoid the penalty with necessary diagrams. (7 Marks)

OR

A,CO3 12. b) i. Assume that 30 percent of the dynamic count of the instructions executed for a program is branch instructions. Delayed branching is used, with one delay slot. Assume that there are no stalls caused by other factors.

A. Derive an expression for the execution time in cycles

- All delay slots are filled with NOP instructions.
- 70 percent of delay slots filled with useful instructions by the optimizing compiler.

B. Determine the compiler's contribution to the increase in performance, expressed as a speedup percentage. (8 Marks)

U,CO3 12. b) ii. Explain in detail about Flynn Classification for parallel architectures with necessary examples. (7 Marks)

A,CO4 13. a) i. Assume a computer system with 32-bits wide physical addresses, 256 K words of cache and four words in a cache block. Show how the 32-bit address would be partitioned for each of the following cache configurations:

A) Direct mapped

B) 2-way set associative

C) 4-way set associative

D) Fully associative (8 Marks)

U,CO4 13. a) ii. Explain in detail about the role of TLB in Virtual Memory with suitable diagrams. (7 Marks)

OR

A,CO4 13. b) i. Construct a larger memory of size 2M x 32 using smaller memory chips of size 256 K x 8 and explain (8 Marks)

U,CO4 13. b) ii. Write in detail about the characteristics of Optical Disk with suitable diagrams. (7 Marks)

U,CO5 14. a) i. Explain in detail about the data transfer during an input operation using Synchronous Bus with neat diagrams. (8 Marks)

U,CO5 14. a) ii. With neat diagrams, explain the characteristics and structure of USB. (7 Marks)

OR

U,CO5 14. b) i. Write in detail about Serial interface circuits for an Input device with its functions. (8 Marks)

U,CO5 14. b) ii. With necessary steps explain the mechanism for handling interrupts from multiple devices. (7 Marks)

PART C – (1 × 20 = 20 Marks)

A,CO1 15. a) Perform the following operations:

A) Use Booth algorithm to find 45 x 30

B) Use Non-Restoring algorithm to find 30 / 4

C) Represent 1.00010… x 2^{-20} and 0.001001… x 2^{10} using IEEE 754-2008 single precision format and perform addition. (20 Marks)

OR

A,CO1 15. b) Perform the following operations:

A) Use Bit Pair Recoding algorithm to multiply 35 x 63.

B) Use Restoring algorithm to find 29 / 6.

C) Multiply 13 x 11 using Sequential Circuit Multiplication. (20 Marks)